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# UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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## ABSTRACT

Vision-language models (VLMs) have advanced rapidly across a broad spectrum of visual question answering (VQA) benchmarks—spanning visual reasoning, document understanding, and multimodal dialogue—and these gains recur consistently across systems built on disparate base models, alignment architectures, and training corpora. This progress, however, conceals a systematic shortfall: recent evidence indicates that such models lag on the traditional image-classification benchmarks that probe fine-grained visual discrimination. We address this gap directly. Recasting four established fine-grained classification benchmarks as multiple-choice VQA so that each model can be probed in its native answer format, we evaluate a large and representative set of 15 recent VLMs and isolate the factors that account for the disconnect between fine-grained recognition and the capabilities that prevailing VQA benchmarks reward. Two observations anchor the study: fine-grained classification accuracy varies widely among models of comparable general VQA ability, marking it as a distinct axis of visual competence that standard benchmarks fail to capture; and VLMs consistently underperform the very CLIP-style encoders they are built upon, leaving substantial visual capability unrealized. To explain these patterns, we conduct a controlled series of LLaVA-style ablations that hold the training recipe fixed while varying one design factor at a time. Within this setting, upgrading the base language model improves fine-grained and general VQA scores in tandem, whereas a stronger vision encoder improves fine-grained classification selectively; the pretraining stage proves especially consequential for fine-grained performance, particularly when the language-model weights are unfrozen during pretraining, while pretraining-data quality and instruction tuning exert comparatively limited influence. Together, these findings chart a concrete path toward strengthening the fine-grained, vision-centric capabilities of future VLMs.

## 1 INTRODUCTION

Vision-language models (VLMs), which couple vision encoders with large language models (LLMs), have driven rapid progress in interpreting and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b), attaining strong performance on demanding tasks that range from open-ended visual question answering to document understanding and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). This trajectory of success, however, leaves a foundational question largely unexamined: how well do contemporary VLMs perform on fine-grained visual recognition, and which design choices govern that performance? This paper takes up both questions in a single controlled study—first quantifying where present-day VLMs stand on fine-grained recognition relative to their own vision encoders and to prevailing benchmarks, then isolating, within a fixed training recipe, the components that most strongly shape it.

Fine-grained visual recognition—the discrimination of visually similar categories—requires a model to attend to subtle distinguishing features while remaining invariant to intra-class variation in pose, lighting, and background. The traditional vision encoders on which modern VLMs are built, notably CLIP (Radford et al., 2021) and DINO (Caron et al., 2021), attain strong fine-grained classification accuracy; the fine-grained competence that survives their integration into a VLM, however, has gone comparatively unmeasured—an omission this work sets out to correct.

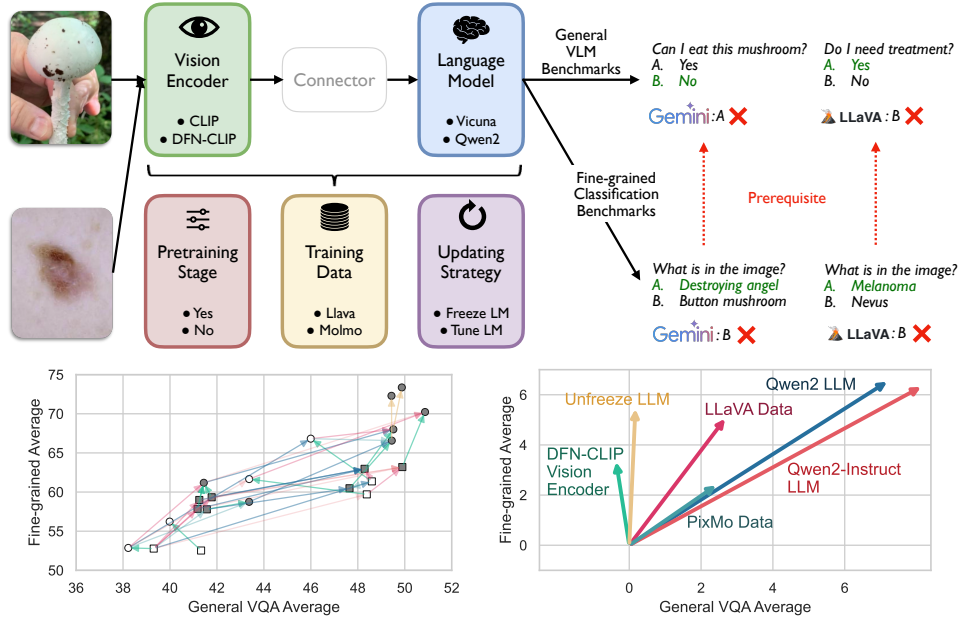


Figure 1: **Overview.** (Top) We characterize the fine-grained classification capabilities of vision-language models (VLMs)—an essential yet routinely overlooked competence that underpins higher-level visual understanding and reasoning. (Bottom) Through 22 systematic, single-factor ablation experiments over key model components and training strategies (left), conducted within a fixed LLaVA-style recipe, we identify the factors that drive fine-grained classification performance in VLMs (right).

Characterizing the fine-grained recognition of VLMs is essential precisely because perception is the substrate on which higher-level understanding and reasoning are built. Real-world deployments routinely confront long-tailed distributions in which discriminating among closely related categories is decisive rather than incidental: a model that cannot correctly identify a mushroom species cannot, in turn, reason from its textual knowledge about whether that species is poisonous or safe to consume. Beyond such immediate stakes, understanding the factors that govern fine-grained knowledge acquisition directly informs the architectures and training strategies of future VLMs.

We first present a broad, head-to-head evaluation of fine-grained visual recognition in state-of-the-art VLMs, juxtaposing each model’s fine-grained classification accuracy with its competence on general VLM tasks. To probe every model in the multiple-choice format on which contemporary VLMs are trained and evaluated, we recast four established fine-grained benchmarks—ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014)—into a common multiple-choice protocol with hard negatives, and evaluate 15 recent VLMs, among them LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024). We set these results against aggregated performance on eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). Two findings emerge. First, models of comparable general ability differ markedly in fine-grained classification accuracy, which establishes fine-grained recognition as a distinct axis of visual intelligence that prevailing VLM benchmarks fail to capture. Second, VLMs consistently trail the CLIP-based vision encoders on which they are built—a gap that corroborates and, across a wider range of models and datasets, extends prior observations (Zhang et al., 2024; Geigle et al., 2024) and reveals substantial fine-grained capability left unrealized.

To explain what drives fine-grained classification performance, we then conduct a controlled study that disentangles the contributions of the principal model components and training paradigms. Holding a single LLaVA-style training recipe fixed, we vary one factor at a time—the language model, the vision encoder, whether pretraining is performed and how, and the quality of pretraining and instruction-tuning data—so that each measured difference is attributable to the ablated factor rather than to incidental implementation changes. Across 22 such single-factor ablations, and within the

scale and recipe of this controlled setting, we identify the factors that most strongly shape fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Upgrading the base language model raises performance across both fine-grained classification and general VLM benchmarks, improving the two in tandem rather than trading one against the other.
- **Vision encoder (§4.2):** A stronger vision encoder improves fine-grained classification selectively, leaving general VLM benchmarks largely unchanged—an asymmetry most pronounced when the encoder is integrated through a two-stage training procedure.
- **Pretraining stage (§4.3):** Pretraining on image-captioning data substantially boosts fine-grained classification, with a comparatively modest effect on general VLM benchmarks.
- **Weight-updating strategy (§4.3):** Jointly updating the vision-LLM connector and the LLM during pretraining yields markedly stronger fine-grained classification than updating the connector alone.
- **Data quality (§4.3):** Within our setup, the quality of pretraining and instruction-tuning data exerts comparatively limited influence on fine-grained classification, particularly when the language-model weights remain frozen during pretraining.

In summary, this study places fine-grained visual recognition—an essential capability that current benchmark evaluations systematically overlook—at the center of VLM analysis, and, through controlled single-factor ablations, identifies the components that most strongly govern fine-grained knowledge acquisition. Taken together, the resulting evidence offers an actionable blueprint for building more vision-centric VLMs, and thereby for improving their reliability on the real-world long-tail tasks that hinge on fine-grained recognition.

## 2 RELATED WORK

**Vision-language models and their evaluation.** Vision-language models (VLMs) integrate visual inputs with language models; most contemporary systems adopt the LLaVA-style architecture (Liu et al., 2023), which comprises three core components—a vision encoder, a language model, and an MLP connector that projects visual features into the language model’s input space. Such models attain strong performance across a wide range of multimodal tasks, spanning visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Their evaluation has been propelled by a proliferation of benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks, however, predominantly probe reasoning and language understanding conditioned on visual inputs, and in doing so they leave core vision-centric capabilities—object recognition and fine-grained classification foremost among them—largely unmeasured. We close this gap by systematically evaluating and analyzing VLMs on fine-grained classification, an essential yet underexplored dimension of visual intelligence.

**Fine-grained visual classification.** Fine-grained classification is a well-established computer vision task that seeks to distinguish visually similar subcategories within a broader object category, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). The associated benchmarks have long served as central testbeds for training and evaluating vision encoders, including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Despite this centrality to vision research, recent VLM evaluation protocols have largely set these datasets aside in favor of visual question answering. Yet fine-grained classification is precisely the capability demanded by real-world, long-tailed applications that hinge on precise recognition—medical diagnosis, food safety, and species identification among them—so evaluating VLMs on such benchmarks is indispensable for assessing their practical, vision-centric usability rather than their language-centric reasoning alone.

**Evaluation and ablation of VLMs for fine-grained classification.** Prior studies of VLM fine-grained classification (Zhang et al., 2024; Geigle et al., 2024) remain narrow in scope, typically confined to a handful of models or datasets. We substantially broaden this line of work, evaluating a considerably wider range of VLMs across four fine-grained classification benchmarks under a single unified multiple-choice protocol. Crucially, we move beyond evaluation alone to a controlled

ablation analysis that isolates the design factors governing fine-grained classification performance within a fixed training recipe.

A parallel body of work has probed the VLM design space, examining model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). These analyses, however, center on general, language- and reasoning-oriented benchmarks. Our study instead traces how the pivotal design choices—base model selection, pretraining methodology, and the composition of pre-training and instruction-tuning data—bear on fine-grained classification specifically, yielding guidance for optimizing VLMs on vision-centric tasks that complements this prior, reasoning-oriented analysis.

### 3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates the study of fine-grained classification in VLMs, specifies our benchmarks and evaluation protocol, and reports the key findings from testing 15 existing VLMs.

#### 3.1 MOTIVATION

Fine-grained classification is a fundamental computer vision task that requires distinguishing visually similar subcategories within a broader category, such as individual bird species or car models (Wei et al., 2021). A substantial suite of benchmarks, ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) among them, has been established as the standard testbed for training and evaluating vision encoders. Driven by large-scale self-supervised and supervised learning, encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) now attain near- or super-human accuracy on these challenging fine-grained tasks.

The evaluation of vision-language models (VLMs), by contrast, has largely set these benchmarks aside. Although recent VLMs such as LLaVA (Liu et al., 2023) interpret visual concepts effectively, their evaluation is typically confined to visual question answering (VQA)—academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document-understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Because these benchmarks foreground reasoning and language processing while omitting fine-grained object categories, they leave core vision-centric capabilities largely unexamined.

Fine-grained classification is nonetheless consequential for VLMs, because real-world applications routinely encounter long-tailed distributions that demand precise discrimination among closely related categories, and the cost of error is often severe. Consider a visually impaired user who asks whether a mushroom is safe to eat: the model must first identify the species correctly before it can reason about toxicity, and mistaking a deadly destroying angel for an edible white button mushroom could prove fatal. Medical diagnosis is analogous—failing to distinguish visually similar conditions can precipitate incorrect treatment and serious harm—as is autonomous driving, where confusing a “stop” sign with a “do not enter” sign, both red and similarly shaped, could produce dangerous navigation errors. Competence at fine-grained classification is, in short, a prerequisite for the safety and reliability of VLMs in real-world deployment.

#### 3.2 BENCHMARKS

We assess the fine-grained visual recognition of VLMs using four well-established object-recognition benchmarks, each targeting a distinct domain and together spanning generic objects, flowers, pets, and food: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

Because these benchmarks span 37 to 1,000 distinct classes, presenting the full label set as answer options is impractical within the multiple-choice format on which contemporary VLMs are trained and evaluated. We therefore adopt the conversion methodology of Geigle et al. (2024) to recast each dataset into 5-way multiple-choice questions. For every image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negatives by image-label cosine similarity, which are then combined with the ground-truth label to form the answer set. We deliberately mine hard negatives with an OpenCLIP model, rather than any encoder embedded in an evaluated

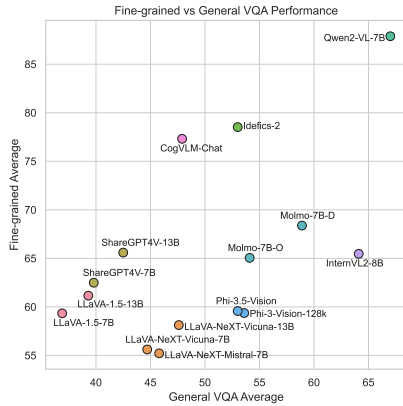


Figure 2: **Fine-grained classification versus general VQA performance across VLMs.** Fine-grained classification accuracy (averaged over our four benchmarks) varies widely among recent VLMs of comparable general VQA ability (averaged over eight benchmarks), establishing it as a distinct axis of visual capability that standard VQA benchmarks fail to measure.

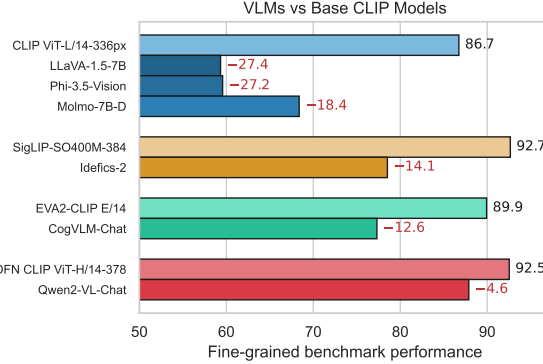


Figure 3: **VLMs versus their corresponding CLIP vision encoders on fine-grained classification.** Each VLM is compared, in our multiple-choice protocol, against the zero-shot accuracy of the CLIP encoder it is built upon. Only Qwen2-VL-Chat approaches its DFN-CLIP encoder; every other VLM trails its own encoder by a substantial margin, indicating that current VLMs leave much of their encoders’ fine-grained capability unrealized.

VLM, so as not to bias difficulty against VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), each trained on a different dataset than the OpenCLIP model used for mining. This conversion probes each VLM in its native answer format while preserving most of the underlying difficulty: CLIP ViT-L/14-336px (Radford et al., 2021), for instance, attains 75.2% accuracy on the full 102-class Flowers-102 and 78.3% on our 5-way multiple-choice version, confirming that the reformulation—rather than trivializing recognition—leaves the discrimination substantively hard.

### 3.3 TESTING SETUP

We evaluate 15 recent VLMs spanning the 7B–13B parameter range on the converted 5-way fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). To respect the answer conventions under which each model was tuned, we adopt its default prompt format from VLMEvalKit (Duan et al., 2024) and score accuracy by exact match against the multiple-choice options. As a reference upper bound, we evaluate a range of CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) in the standard zero-shot setup, encoding the image and each class label—instantiated in the template “A photo of a [class]”—with CLIP’s image and text encoders and assigning the top-1 prediction by maximum image–text cosine similarity.

To determine whether existing VLM evaluation benchmarks already capture fine-grained classification ability, we contrast each VLM’s fine-grained accuracy with its average score across eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

### 3.4 RESULTS

Figure 2 plots fine-grained classification performance (averaged over our four benchmarks) against general VQA performance (averaged over eight existing VQA benchmarks) for every tested VLM. The two dimensions decouple sharply: models with nearly identical general VQA scores span a wide range of fine-grained classification accuracy. CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a), for instance, both reach roughly 48% average accuracy on

general VQA benchmarks, yet their fine-grained classification accuracy differs by 19 percentage points (pp)—77.3% for CogVLM-Chat against 58.1% for LLaVA-NeXT-Vicuna-13B. A gap this large at matched general ability establishes that prevailing VLM benchmarks do not capture fine-grained classification, which constitutes instead a distinct dimension of visual intelligence.

**Finding 1:** Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Because every current VLM is built on a pretrained vision encoder, our multiple-choice format affords a direct, like-for-like comparison between each VLM and the CLIP model it embeds on traditional classification benchmarks; we acknowledge that the two are probed under different protocols—multiple-choice prompting for the VLM against zero-shot cosine matching for the encoder—so the encoder score is best read as a reference point for the capability latent in the shared backbone rather than as a strictly matched control. Figure 3 sets the fine-grained accuracy of selected VLMs against their corresponding CLIP baselines and exposes a consistent, substantial gap. Molmo (Deitke et al., 2024), a very recent model that excels on general VQA benchmarks, attains only 68.4% average accuracy in our multiple-choice setting, whereas its own vision encoder, CLIP ViT-L/14 (Radford et al., 2021), reaches 86.7%. Even the strongest model we test, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) encoder by 4.6 points. This systematic shortfall relative to the very encoders VLMs contain reveals substantial fine-grained capability left untapped, and motivates a controlled analysis of the design choices that could unlock it. The next section undertakes precisely this ablation study.

**Finding 2:** A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

## 4 ABLATIONS: BASE MODELS AND TRAINING

### 4.1 EXPERIMENTAL SETUP

Building on the observational results of Section 3, we ablate the model-level factors that plausibly account for fine-grained classification performance, concentrating on base model selection and training methodology. All experiments are conducted within the single LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023), which we hold fixed so that each measured difference is attributable to the ablated factor rather than to incidental implementation changes. Each model undergoes one epoch of pretraining (where applicable) followed by one epoch of finetuning, with LLaVA-1.5’s default hyperparameters—learning rate and batch size included—retained throughout. Because this fixed recipe operates at a deliberately controlled scale, and because we report single training runs rather than averages over seeds, we frame the resulting effects as consistent trends within this setting rather than as recipe-independent laws; the magnitudes should be read accordingly.

Fine-grained performance is reported as the mean accuracy over the four classification benchmarks introduced in Section 3.2. General VQA performance is reported as the mean score over three widely adopted multiple-choice benchmarks—MMM (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b)—chosen for their broad domain coverage and prevalence in VLM evaluation.

### 4.2 BASE MODELS

The analysis in Section 3 reveals a clear trend: recent models routinely adopt newer base LLMs, yet most retain the CLIP ViT-L/14 vision encoder of Radford et al. (2021). We isolate how each of these two base-model choices bears on fine-grained classification, varying one component while holding the other fixed.

#### 4.2.1 LLM CHOICE

**Setup.** Taking LLaVA-1.5-7B (Liu et al., 2023) as the baseline, we ablate the LLM component while varying the vision encoder and pretraining configuration. We contrast the original Vicuna-7B-

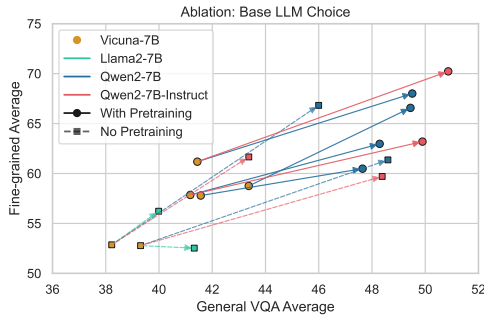


Figure 4: **Base-LLM ablation: replacing LLaVA’s Vicuna-7B with alternative LLMs.** Averaged over vision-encoder and pretraining configurations, switching from Vicuna to Qwen2-7B raises fine-grained performance by +7.5pp and general VQA performance by +7.7pp—improving both dimensions in step rather than trading one for the other.

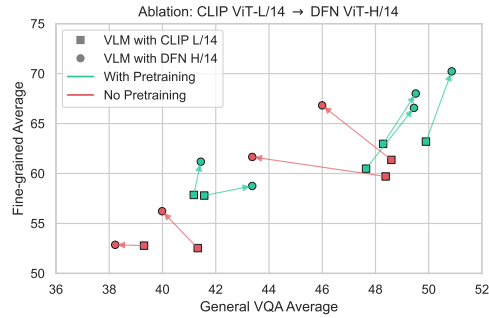


Figure 5: **Base vision-encoder ablation: replacing LLaVA’s CLIP ViT-L/14 with DFN-CLIP ViT-H/14.** Given connector pretraining, switching to DFN-CLIP improves fine-grained performance by +4.5pp while general performance rises only +1.2pp—an asymmetry that isolates the vision encoder’s specifically fine-grained role.

1.5 (Chiang et al., 2023) used in LLaVA with Qwen2-7B (Yang et al., 2024)—the LLM underlying Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)—together with Llama2-7B (Touvron et al., 2023), Vicuna’s base model, and Qwen2-7B-Instruct, the instruction-tuned variant of Qwen2-7B.

**Findings.** Figure 4 shows that replacing Vicuna-7B with Qwen2-7B improves performance substantially and consistently across every configuration, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct yields comparable gains (+7.5pp and +8.1pp, respectively), which indicates that the effect is a property of the stronger base LLM rather than of a particular instruction-tuning recipe. Llama2-7B, the non-instruction-tuned base of Vicuna, yields only modest improvements, leaving the downstream consequences of omitting instruction tuning an open question. Two mechanisms could in principle produce these gains—better visual grounding or better multiple-choice reasoning and instruction following—and our design does not fully disentangle them; one control is nonetheless informative. Every trained model, the weaker Vicuna-based VLMs included, produces correctly formatted multiple-choice responses, so the fine-grained differences reflect genuine discrimination among the presented options rather than mere disparities in format adherence.

**Takeaway 1:** Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

#### 4.2.2 VISION ENCODER CHOICE

**Setup.** Most of the VLMs we evaluate, including the LLaVA series and Molmo, adopt OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. Qwen2-VL (Wang et al., 2024b), by contrast, employs DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which exhibits stronger zero-shot performance on object-recognition tasks such as ImageNet. We evaluate both encoders across the full range of LLM choices and pretraining configurations.

**Findings.** Figure 5 shows that the effect of substituting CLIP L/14 with DFN-CLIP H/14 is contingent on whether connector pretraining precedes finetuning. Absent pretraining, DFN-CLIP slightly degrades general VQA performance (-2.5pp) while modestly improving fine-grained scores (+2.8pp). Once connector pretraining is performed, however, the picture changes: general VQA performance holds steady or improves marginally (+1.2pp), whereas fine-grained performance rises substantially (+4.5pp). A stronger vision encoder thus translates into markedly better fine-grained classification—but only when it is properly integrated through pretraining, and its benefit concentrates on the fine-grained axis rather than diffusing across general ability. This interaction between

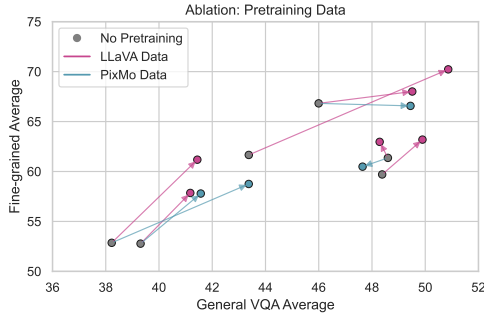


Figure 6: **Pretraining-data ablation.** Starting from no pretraining, adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) connector-pretraining data yields comparable gains: LLaVA data raises fine-grained and general scores by +4.0pp and +2.1pp respectively, PixMo data by +2.4pp and +2.5pp—even though the two sources differ sharply in caption quality (9.8 versus 169 words on average).

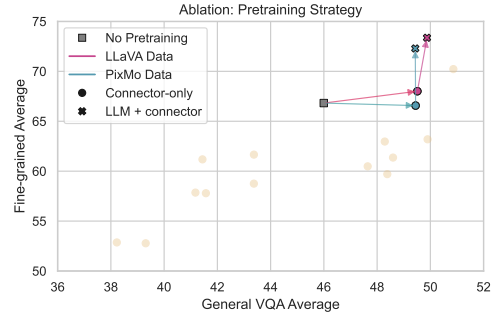


Figure 7: **Weight-updating ablation: connector-only pretraining versus joint LLM+connector tuning.** Unfreezing the LLM during pretraining (after a 20% connector-only warmup) improves fine-grained performance by +5.5pp with no adverse effect on general VQA scores, under both LLaVA and PixMo pretraining data.

the encoder and the pretraining stage indicates that the two factors are not independent, and cautions against reading any single ablation in isolation from the training procedure that surrounds it.

**Takeaway 2:** Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

### 4.3 TRAINING METHODS

Recent VLMs are predominantly trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, optimizes a captioning objective over large-scale image-caption datasets to align the vision encoder with the LLM. The second stage, instruction tuning, refines the VLM on curated multi-round, multimodal instruction-completion data, aligning its outputs with human intent for real-world use. We now quantify how each stage shapes the fine-grained capabilities of VLMs within our controlled recipe.

#### 4.3.1 PRETRAINING

**Setup.** We address three questions about the pretraining stage:

- Is pretraining necessary for LLaVA-architecture models when their base components are already pretrained on billion-scale data?
- How does connector-only pretraining compare with jointly unfreezing the LLM?
- How much does pretraining-data quality matter for downstream performance?

We compare three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, comprising high-quality, human-annotated detailed captions), and no pretraining at all. In most experiments we pretrain the connector alone, following Liu et al. (2023). We additionally examine a joint strategy in the spirit of Deitke et al. (2024): pretraining the connector for the first 20% of steps as a warmup, then unfreezing the LLM for the remaining 80%. Every configuration concludes with one epoch of finetuning on LLaVA finetuning data, so that differences trace to the pretraining manipulation alone.

**Findings.** Figure 6 presents the pretraining ablation. Connector-only pretraining improves VLM performance in nearly every setting, the sole exception being the Qwen2 and CLIP ViT-L/14 combination. The gains are most pronounced for Vicuna-based VLMs, which improve on average by

Table 1: **Summary of the key ablation settings, with existing reference models in bold.** Within our fixed LLaVA-style recipe, upgrading the base LLM and vision encoder, together with pretraining the connector and LLM on sufficient data, drives the gains in fine-grained performance, whereas instruction finetuning contributes comparatively little; parenthetical values report the change relative to the preceding row. General VQA benchmarks follow a distinct pattern, in which the choice of LLM plays the larger role.

| Model/Ablation            | Vision Encoder | LLM    | Arch.    | Pretraining Data | Finetuning Data | Fine-grained Classification | General VQA |
|---------------------------|----------------|--------|----------|------------------|-----------------|-----------------------------|-------------|
| <b>LLaVA-1.5-7B</b>       | CLIP L/14      | Vicuna | LLaVA    | LLaVA            | LLaVA           | 59.3                        | 41.8        |
| No pretraining            | CLIP L/14      | Vicuna | LLaVA    | None             | LLaVA           | 52.8                        | 39.3        |
| LLaVA reproduction        | CLIP L/14      | Vicuna | LLaVA    | LLaVA            | LLaVA           | 57.9 (+5.1)                 | 41.2 (+1.9) |
| Qwen2 LLM                 | CLIP L/14      | Qwen2  | LLaVA    | LLaVA            | LLaVA           | 63.0 (+5.1)                 | 48.3 (+7.1) |
| DFN-CLIP encoder          | DFN H/14       | Qwen2  | LLaVA    | LLaVA            | LLaVA           | 68.0 (+5.0)                 | 49.5 (+1.2) |
| Unfreeze LLM              | DFN H/14       | Qwen2  | LLaVA    | LLaVA            | LLaVA           | 73.4 (+5.4)                 | 49.9 (+0.4) |
| Pretrain on PixMo         | CLIP L/14      | Qwen2  | LLaVA    | PixMo            | LLaVA           | 66.6                        | 49.4        |
| <b>Molmo-7B-D</b>         | CLIP L/14      | Qwen2  | Molmo    | PixMo            | Molmo           | 68.4                        | 58.0        |
| FT Qwen2-VL base on LLaVA | DFN H/14       | Qwen2  | Qwen2-VL | Qwen2-VL         | LLaVA           | 85.5                        | 61.1        |
| <b>Qwen2-VL-7B</b>        | DFN H/14       | Qwen2  | Qwen2-VL | Qwen2-VL         | Qwen2-VL        | 87.9 (+2.4)                 | 62.4 (+1.3) |

+6.1pp on fine-grained classification and +3.1pp on general VQA—indicating that, even when both base components arrive pretrained on billion-scale data, the alignment pretraining stage remains a substantial lever for fine-grained recognition.

**Takeaway 3:** Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Strikingly, pretraining on low-quality web-scraped captions (LLaVA) and on highly detailed human annotations (PixMo) yields nearly indistinguishable outcomes: relative to LLaVA data, PixMo data produces only a -1.6pp change in fine-grained performance and a +0.4pp change in general VQA scores, despite the two caption sources differing by more than an order of magnitude in length and richness.

**Takeaway 4:** Pretraining data quality has a limited impact on overall model performance.

We hypothesize that connector-only pretraining prevents the model from fully exploiting higher-quality captions, because the frozen LLM cannot absorb the richer supervisory signal they carry. To test this hypothesis, we pretrain the LLM together with the connector, prefaced by the 20% connector-only warmup described above. Figure 7 confirms that this joint scheme substantially improves fine-grained classification (+5.5pp) while leaving general VQA scores intact. The gain nonetheless materializes under both LLaVA and PixMo pretraining data, which indicates that unfreezing the LLM—not the higher caption quality per se—drives the improvement, and that pretraining on lower-quality captions does not erode the LLM’s language capabilities.

**Takeaway 5:** Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

### 4.3.2 FINETUNING

**Setup.** Departing from the preceding protocol, we start from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b)—the model from which Qwen2-VL-7B-Chat is derived—and finetune it on LLaVA-Instruct data (Liu et al., 2023) using our earlier procedure. We benchmark the result against Qwen2-VL-7B-Chat, which is instruction-finetuned on 2M in-house examples (Wang et al., 2024b), thereby isolating the effect of the finetuning dataset while holding the pretrained base fixed.

**Findings.** Finetuning on LLaVA data alone slightly lowers both fine-grained and general benchmark performance: fine-grained scores fall by 2.4pp (from 87.9% to 85.5%) and general scores by 1.3pp (from 62.4% to 61.1%). As Figure 8 makes clear, this is the smallest fine-grained contribution among all our ablations, which places instruction finetuning in a distinctly secondary role for fine-grained

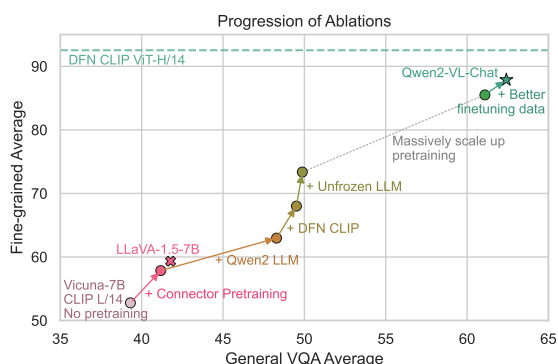


Figure 8: **Cumulative effect of the combined ablations.** Beginning from Vicuna-7B with CLIP ViT-L/14, progressively upgrading the base LLM, the vision encoder, and the training procedure raises fine-grained classification accuracy from 52.8% to 73.4%, closing more than half the gap to Qwen2-VL-Chat’s 87.9%. The residual difference is most plausibly attributable to Qwen2-VL’s far larger pretraining corpus of 1.4T tokens, a scale our controlled runs do not reach.

knowledge acquisition relative to base-model selection and pretraining strategy—at least when the finetuning corpora compared are of the scale examined here.

**Takeaway 6:** The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

#### 4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize our ablations, tracing how each component narrows the performance gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) with CLIP ViT-L/14 (Radford et al., 2021)), our sequence of modifications lifts fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The decomposition is clean and interpretable: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately strengthen fine-grained capabilities relative to general ability, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for most of the gain in general VQA. Each factor thus maps onto the dimension our component-wise ablations predicted it would drive, a coherence that lends the decomposition credence even though it reflects a single ordered sequence rather than a full factorial design.

These interventions notwithstanding, a residual 12-point gap in fine-grained classification remains beyond the reach of the components we ablate (Table 1). Two candidate explanations present themselves: architectural differences and pretraining-data scale.

Of the two, the available evidence favors pretraining-data scale over architecture. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) experiments use comparatively small corpora—each fewer than 1M image-caption pairs, roughly 200M and 400M tokens respectively—whereas Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, orders of magnitude more. This disparity, reinforced by our observational results (Section 3.4) and training ablations (Section 4.3), most parsimoniously accounts for the remaining gap and identifies extensive pretraining as the critical, and as yet unexhausted, lever for developing VLMs with superior fine-grained classification. Directly confirming this attribution at the trillion-token scale lies beyond our compute budget and constitutes the most consequential direction for future work.

## 5 LIMITATIONS & CONCLUSION

Our study has practical limitations that in turn delineate concrete directions for future work. Under computational constraints, our controlled training compares configurations at the <1M data-point scale rather than the billion-scale (B) regime used by some newer VLMs, which leaves open how

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the observed effects extrapolate as pretraining data grows; this axis of scale is, in our view, the most consequential to explore next. Our causal claims are correspondingly bounded to a single LLaVA-style recipe evaluated without a full factorial design or repeated seeds, so the reported magnitudes are best read as consistent within-recipe trends rather than as recipe-independent constants. Our evaluation, moreover, probes fine-grained recognition specifically through a multiple-choice protocol with OpenCLIP-mined hard negatives; complementary formats—open-vocabulary prediction, alternative hard-negative sources, and larger candidate sets—together with a closer accounting of possible benchmark–pretraining overlap for these long-standing datasets, remain natural avenues for corroborating and extending our conclusions. More recent work also proposes alternative pretraining strategies whose interaction with fine-grained visual understanding warrants dedicated study.

In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks and establish fine-grained visual recognition as a crucial yet underexplored dimension of VLM capability—one that prevailing VQA benchmarks fail to measure and on which VLMs consistently trail their own vision encoders. Through a controlled, single-factor analysis of the principal model components and training paradigms, we identify actionable strategies—stronger language models, better-integrated vision encoders, and, above all, adequate pretraining—for improving fine-grained classification and strengthening vision-centric competence, and thereby for advancing the reliability of VLMs in the real-world scenarios that demand precise visual understanding.

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| Vision Encoder    | LLM               | Arch     | PT Data  | FT Data  | Updating  | Classification | VLM  |
|-------------------|-------------------|----------|----------|----------|-----------|----------------|------|
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 59.3           | 41.8 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 59.0           | 41.2 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | None     | LLaVA    | Connector | 52.8           | 39.3 |
| CLIP ViT-L/14@336 | Llama2-7B         | LLaVA    | None     | LLaVA    | Connector | 52.5           | 41.3 |
| DFN ViT-H/14@378  | Llama2-7B         | LLaVA    | None     | LLaVA    | Connector | 56.2           | 40.0 |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | None     | LLaVA    | Connector | 52.9           | 38.2 |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | None     | LLaVA    | Connector | 61.4           | 48.6 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | None     | LLaVA    | Connector | 66.8           | 46.0 |
| CLIP ViT-L/14@336 | Qwen2-7B-Instruct | LLaVA    | None     | LLaVA    | Connector | 59.7           | 48.4 |
| DFN ViT-H/14@378  | Qwen2-7B-Instruct | LLaVA    | None     | LLaVA    | Connector | 61.7           | 43.4 |
| DFN ViT-H/14@378  | Qwen2-7B          | Qwen2-VL | Qwen2-VL | LLaVA    | Full      | 85.5           | 61.1 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 57.9           | 41.2 |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 61.2           | 41.4 |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Connector | 63.0           | 48.3 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Connector | 68.0           | 49.5 |
| CLIP ViT-L/14@336 | Qwen2-7B-Instruct | LLaVA    | LLaVA    | LLaVA    | Connector | 63.2           | 49.9 |
| DFN ViT-H/14@378  | Qwen2-7B-Instruct | LLaVA    | LLaVA    | LLaVA    | Connector | 70.2           | 50.9 |
| DFN ViT-H/14@378  | Qwen2-7B          | Qwen2-VL | Qwen2-VL | Qwen2-VL | Full      | 87.9           | 62.4 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | PixMo    | LLaVA    | Connector | 57.8           | 41.6 |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | PixMo    | LLaVA    | Connector | 58.8           | 43.4 |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Connector | 60.5           | 47.6 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Connector | 66.6           | 49.4 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Full      | 73.4           | 49.9 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Full      | 72.3           | 49.4 |

Table 2: **Detailed model-ablation results (aggregated).** Each row reports fine-grained classification and general VQA accuracy for one configuration of vision encoder, LLM, architecture, pre-training data, finetuning data, and weight-updating scheme, providing the complete basis for the comparisons summarized in the main text.

| Vision Encoder    | LLM               | Arch     | PT Data  | FT Data  | Updating  | MMBench | MMMU | MMStar |
|-------------------|-------------------|----------|----------|----------|-----------|---------|------|--------|
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 60.4    | 32.2 | 32.7   |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 58.6    | 32.9 | 32.3   |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | None     | LLaVA    | Connector | 54.4    | 32.8 | 30.7   |
| CLIP ViT-L/14@336 | Llama2-7B         | LLaVA    | None     | LLaVA    | Connector | 58.0    | 34.3 | 31.7   |
| DFN ViT-H/14@378  | Llama2-7B         | LLaVA    | None     | LLaVA    | Connector | 53.9    | 34.6 | 31.5   |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | None     | LLaVA    | Connector | 51.3    | 33.2 | 30.1   |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | None     | LLaVA    | Connector | 63.1    | 41.8 | 40.9   |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | None     | LLaVA    | Connector | 61.3    | 36.2 | 40.5   |
| CLIP ViT-L/14@336 | Qwen2-7B-Instruct | LLaVA    | None     | LLaVA    | Connector | 63.2    | 41.4 | 40.5   |
| DFN ViT-H/14@378  | Qwen2-7B-Instruct | LLaVA    | None     | LLaVA    | Connector | 55.7    | 38.3 | 36.1   |
| DFN ViT-H/14@378  | Qwen2-7B          | Qwen2-VL | Qwen2-VL | LLaVA    | Full      | 78.5    | 49.0 | 55.8   |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 60.4    | 29.9 | 33.3   |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 57.4    | 34.2 | 32.7   |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Connector | 64.4    | 41.3 | 39.1   |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Connector | 64.9    | 41.3 | 42.3   |
| CLIP ViT-L/14@336 | Qwen2-7B-Instruct | LLaVA    | LLaVA    | LLaVA    | Connector | 66.5    | 41.3 | 41.9   |
| DFN ViT-H/14@378  | Qwen2-7B-Instruct | LLaVA    | LLaVA    | LLaVA    | Connector | 68.0    | 41.6 | 43.0   |
| DFN ViT-H/14@378  | Qwen2-7B          | Qwen2-VL | Qwen2-VL | Qwen2-VL | Full      | 78.9    | 49.9 | 58.4   |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | PixMo    | LLaVA    | Connector | 57.8    | 32.6 | 34.3   |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | PixMo    | LLaVA    | Connector | 61.1    | 34.1 | 34.9   |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Connector | 64.9    | 40.0 | 38.0   |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Connector | 63.0    | 41.0 | 44.3   |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Full      | 66.0    | 42.0 | 41.6   |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Full      | 63.7    | 40.0 | 44.6   |

Table 3: **Detailed model-ablation results on the individual general VQA benchmarks,** reporting per-benchmark accuracy on MMBench, MMMU, and MMStar for each configuration; the aggregated general VQA column of Table 2 is their mean.

## A FULL RESULTS

For completeness, and to make every comparison in the main text fully reproducible, we report the full ablation results—aggregated performance, per-benchmark general VQA scores, and per-benchmark fine-grained classification scores—in Table 2, Table 3, and Table 4, respectively.

| Vision Encoder    | LLM               | Arch     | PT Data  | FT Data  | Updating  | ImageNet | Flowers | Pets | Food |
|-------------------|-------------------|----------|----------|----------|-----------|----------|---------|------|------|
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 70.9     | 50.7    | 46.3 | 69.4 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 68.7     | 50.4    | 46.5 | 70.3 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | None     | LLaVA    | Connector | 64.3     | 46.1    | 40.4 | 60.4 |
| CLIP ViT-L/14@336 | Llama2-7B         | LLaVA    | None     | LLaVA    | Connector | 64.4     | 44.2    | 40.7 | 60.8 |
| DFN ViT-H/14@378  | Llama2-7B         | LLaVA    | None     | LLaVA    | Connector | 65.4     | 50.5    | 46.2 | 62.8 |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | None     | LLaVA    | Connector | 62.3     | 48.7    | 39.2 | 61.3 |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | None     | LLaVA    | Connector | 70.6     | 44.1    | 55.6 | 75.2 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | None     | LLaVA    | Connector | 74.1     | 50.8    | 62.3 | 80.1 |
| CLIP ViT-L/14@336 | Qwen2-7B-Instruct | LLaVA    | None     | LLaVA    | Connector | 71.4     | 39.6    | 53.1 | 74.7 |
| DFN ViT-H/14@378  | Qwen2-7B-Instruct | LLaVA    | None     | LLaVA    | Connector | 72.0     | 46.0    | 54.1 | 74.5 |
| DFN ViT-H/14@378  | Qwen2-7B          | Qwen2-VL | Qwen2-VL | LLaVA    | Full      | 84.8     | 79.5    | 87.1 | 90.7 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 69.1     | 50.0    | 45.3 | 67.1 |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | LLaVA    | LLaVA    | Connector | 72.7     | 53.2    | 48.5 | 70.4 |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Connector | 71.5     | 46.9    | 56.6 | 76.9 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Connector | 76.1     | 54.1    | 59.1 | 82.7 |
| CLIP ViT-L/14@336 | Qwen2-7B-Instruct | LLaVA    | LLaVA    | LLaVA    | Connector | 74.5     | 45.8    | 55.1 | 77.4 |
| DFN ViT-H/14@378  | Qwen2-7B-Instruct | LLaVA    | LLaVA    | LLaVA    | Connector | 78.5     | 52.3    | 65.4 | 84.7 |
| DFN ViT-H/14@378  | Qwen2-7B          | Qwen2-VL | Qwen2-VL | Qwen2-VL | Full      | 85.9     | 82.3    | 91.0 | 92.3 |
| CLIP ViT-L/14@336 | Vicuna-7B         | LLaVA    | PixMo    | LLaVA    | Connector | 69.2     | 47.4    | 46.7 | 67.8 |
| DFN ViT-H/14@378  | Vicuna-7B         | LLaVA    | PixMo    | LLaVA    | Connector | 70.8     | 50.9    | 44.8 | 68.5 |
| CLIP ViT-L/14@336 | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Connector | 70.8     | 43.7    | 53.9 | 73.6 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Connector | 76.1     | 50.3    | 57.8 | 82.1 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | LLaVA    | LLaVA    | Full      | 79.4     | 60.7    | 67.3 | 86.0 |
| DFN ViT-H/14@378  | Qwen2-7B          | LLaVA    | PixMo    | LLaVA    | Full      | 80.0     | 56.4    | 65.4 | 87.4 |

Table 4: **Detailed model-ablation results on the individual fine-grained classification benchmarks**, reporting per-benchmark accuracy on ImageNet, Flowers, Pets, and Food for each configuration; the aggregated fine-grained column of Table 2 is their mean.

## B EVALUATION

### B.1 FINE-GRAINED CLASSIFICATION

As introduced in Section 3.2, we evaluate VLMs and base vision encoders on four traditional classification benchmarks that jointly span generic objects, flowers, pets, and food:

**ImageNet-1K** (Deng et al., 2009): A foundational image-classification dataset organized by the WordNet hierarchy, spanning a broad set of supercategories, each containing multiple fine-grained subcategories. The ImageNet test set comprises 50,000 images across 1,000 classes.

The WordNet hierarchy underlying ImageNet, however, renders many classes too visually distinct from the remainder to support hard multiple-choice construction—the “person” category, for instance, contains only three classes. To circumvent this limitation, and given the size of the full test set, we draw on the human-annotated incorrect model predictions of Recht et al. (2019) to select images for which hard negative answer choices can in fact be constructed. From an initial pool of 19,056 annotated examples we retain 5,457 images spanning 928 of the original 1,000 classes. Because this curation leaves the surviving classes unevenly represented, we report mean-per-class accuracy for ImageNet.

**Oxford Flowers-102** (Nilsback & Zisserman, 2008): A dataset of 102 flower species marked by high intra-class variation and pronounced inter-class similarity. The Flowers test set comprises 6,149 examples.

**Oxford-IIIT Pet-37** (Parkhi et al., 2012): A collection of 37 pet categories exhibiting challenging variation in pose, lighting, and occlusion. The Pets test set comprises 3,669 examples.

**Food-101** (Bossard et al., 2014): A dataset of 101 food categories characterized by substantial visual diversity within each class. The Food test set comprises 25,250 examples.

**Question generation.** For every fine-grained classification test set except ImageNet (handled as described above), we construct hard negatives by running zero-shot classification on each test example with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We retain the top four incorrect predictions and shuffle them together with the correct label

to form the five-option answer set, so that the distractors are precisely the categories most easily confused with the ground truth.

**Prompt formatting.** In our multiple-choice VQA format, the question is held fixed as “What type of object is in this photo?”, and the answer choices are rendered as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because different VLMs are trained with different multiple-choice prompt formats, we adopt each model’s corresponding format from VLMEvalKit (Duan et al., 2024)—for instance, “Answer with the option’s letter from the given choices directly.”—so that each model is queried under the answer convention it was tuned to follow.

For CLIP models we follow the standard zero-shot evaluation procedure, under which no question format applies; the class names passed to the text encoder are instead instantiated in the template “a photo of a {classname}”.

## B.2 GENERAL VQA

For the initial observational study in Section 3, we rely directly on the scores reported on the Open-VLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average over eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kemahavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This aggregate score enters our analysis only in Figure 2.

For all subsequent evaluations—of both existing models and the models we train in Section 4—we restrict attention to a subset of multiple-choice VLM benchmarks that affords a more direct, controlled comparison. Concretely, we evaluate on three benchmarks:

**MMBench** (Liu et al., 2024b): A dataset of 2,948 examples probing a broad range of VLM capabilities partitioned into perception and reasoning. Although most examples concern coarse-grained perception and visual reasoning, a subset targets “fine-grained perception”; these categories—action recognition, attribute recognition, and OCR among them—occupy a different domain from the fine-grained object-classification datasets we study, and should not be conflated with them.

**MMMU (Val)** (Yue et al., 2024): A large-scale multimodal dataset spanning a wide distribution of domains and applications. While the test set contains 10,500 examples, most work reports performance on the 900-question validation split, a convention we follow. MMMU probes general knowledge tied to visual perception across art, science, engineering, medicine, business, and the humanities.

**MMStar** (Chen et al., 2024b): A dataset of 1,500 human-selected questions, curated to span a diverse set of tasks across both domain and low-level model capabilities.

**Evaluation procedure.** For the general VQA benchmarks we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). Existing models each carry a predefined evaluation procedure; for our own trained models we apply the LLaVA evaluation code, consistent with our exclusive use of LLaVA-Instruct finetuning data.

## C TRAINING

Our implementation builds on the LLaVA-1.5 training codebase (Liu et al., 2023), and for the majority of experiments we adhere closely to the LLaVA architecture and training procedure so that the ablated factor is the only intended source of variation.

**Pretraining.** Pretraining consists of one epoch on either the LLaVA pretraining data, drawn from CC-3M images and captions, or the Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we use only the “caption” field, not the raw audio transcript. Pretraining uses a batch size of 256. In most pretraining runs we update only the randomly initialized MLP connector, at a learning rate of  $1e-3$ . When ablating the pretraining strategy, we first update the connector alone as a warmup for 20% of the steps and then unfreeze the LLM, updating it as well at a learning rate of  $2e-5$  for the remaining 80% of the steps.

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**Finetuning.** Finetuning always runs for one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from a range of datasets to support proper benchmarking. During finetuning we hold the LLM learning rate at  $2e-5$  with a batch size of 128. The vision encoder remains frozen throughout both training phases.

For the ablation on the Qwen2-VL architecture, we initialize from Qwen2-VL-7B Base and train with the same procedure, unfreezing only the LLM.

**Compute.** All models are trained on 4 nodes of 4 A100 (40GB) GPUs each. Training typically completes within 20 hours, the duration scaling with the size of the CLIP model.

**LLaVA vs. PixMo data.** The main text contrasts the caption quality of the LLaVA and PixMo pretraining data; we make that contrast concrete here. The LLaVA pretraining data is sourced from CC-3M and consists of web-scraped captions widely recognized as low in quality and relevance. PixMo-Cap, by contrast, is assembled from human annotators who record a 60–90 second spoken description of an image, which is then consolidated into a coherent caption (Deitke et al., 2024). A surface analysis quantifies the gap: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words—a difference of more than an order of magnitude that makes the near-parity of their downstream effects all the more notable.

*LLaVA caption example:* “front panel bracket cover for suzuki”

*PixMo caption example:* “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”